An ML Emulator of FLEXPART's Convection Scheme

Phase 1 of a hybrid physics–machine-learning Lagrangian dispersion model

Muhammad Waqas

November 2026

Abstract

FLEXPART moves particles vertically in convective clouds with the Emanuel–Živković-Rothman scheme. This report describes Phase 1 of a hybrid model in which FLEXPART's simple equations stay as exact physics and only complex modules are replaced by small trained models. FLEXPART 11's convection routines (`CONVECT`, `TLIFT`) were extracted, patched and compiled as a standalone library that labels atmospheric columns with the scheme's redistribution matrix and new cloud-base mass flux. Four small neural networks (a convection gate, a cloud-top classifier, a mass-conserving redistribution network and a cloud-base mass-flux regressor; ≈1.9 MB in total) were trained on ERA5 columns of the tropical Indian Ocean (2019–2021). In the final version (v6), which uses real surface pressure, they reproduce the mean upward mass-flux profile of the unseen year 2023 within 4.7% at every significant level, conserve mass exactly, and reach a median per-column error of 29% in the moved air. On the tropical Atlantic, a region never seen in training, the per-column structure generalises (errors within 1.12 times the Indian Ocean values), but the mass-flux profile is overestimated by 6–9% above the boundary layer. A diagnostic traces this to a shift in convection regime (shallower convection under higher surface pressure), not to the surface-pressure input, which points to more varied training regions. The emulator is only 1.3–2.4 times faster than the Fortran scheme on a CPU, so the case for the approach rests on accuracy and on learning better physics in later phases, not on speed.

1 Motivation

FLEXPART (Stohl et al., 2005; Pisso et al., 2019; Bakels et al., 2024) is a chain of physical steps: interpolation of meteorological fields, particle release, turbulence statistics, the Langevin equation, advection, convection, removal processes and concentration sampling. Most steps are short closed-form formulas, which a neural network cannot improve on. A few are expensive or empirical: the convection scheme (Emanuel and Živković-Rothman, 1999; Forster et al., 2007), the turbulence statistics and wet scavenging. The overall plan (Table 1) replaces only those modules with trained models, keeping the particle framework, advection and mass bookkeeping as exact physics. Convection is the first target: it is the heaviest physics module per time step, it decides how much ocean-emitted gas (e.g. bromoform) is lifted towards the tropopause, and its mapping from a column profile to a redistribution matrix is deterministic. The approach follows the idea of learned sub-grid parameterisations in climate models (Rasp et al., 2018); ML emulators of whole dispersion-model footprints exist (Fillola et al., 2023; He et al., 2025), but they bypass the physics instead of replacing single modules.

2 Methods

2.1 FLEXPART's convection scheme as the teacher

For each column, FLEXPART fills module arrays with temperature T, specific humidity q, saturation humidity q_s and full- and half-level pressures, and calls `CONVECT`. The scheme returns the

Phase	Work	Status
1	Offline emulator of FLEXPART's convection scheme, trained on FLEXPART's own code	complete (v6)
2	Coupling into FLEXPART (TorchScript model called from Fortran via FTorch); long online runs	next
3	ML turbulence and wet-scavenging modules trained on LES or observations, aiming to beat the original fits	planned
4	Validation of the hybrid model against ship and aircraft observations	planned

Table 1: Project phases.

mass flux f_{ij} ($\mathrm{kg\,m^{-2}\,s^{-1}}$) from level i to level j and updates the cloud-base mass flux m_b, which must be carried between calls: the scheme has memory. FLEXPART's `calcmatrix` turns f_{ij} into a mass matrix; dividing each row by the layer mass gives the redistribution matrix of fractions used here,

$$M_{ij} = \frac{\Delta t\, f_{ij} + \delta_{ij}\left(\dfrac{\Delta p_i}{g} - \Delta t \sum_k f_{ik}\right)}{\Delta p_i / g}, \qquad \sum_j M_{ij} = 1, \tag{1}$$

where Δp_i is the pressure thickness of level i and $\Delta t = 900$ s.

`CONVECT` and `TLIFT` (lines 990–2106 of `conv_mod.f90`, FLEXPART 11) depend only on FLEXPART's parameter module, so they were copied into a standalone Fortran module with the arrays they use and a C-compatible wrapper, compiled with `gfortran`, and called from Python through `ctypes`. One change was needed: the scheme stops the program (`ERROR STOP`, "Convection reaches top level of input data") when convection approaches the top of the column, which killed the Python session. The statement was replaced by a flag, and such columns are returned with `iflag` $= -1$ and dropped. Columns extend to 10 hPa, which made this case rare (0.1% of columns in a synthetic test). In tests, the scheme behaved as expected: convective columns gave `iflag` $= 1$ with m_b spinning up over successive calls (0.0029 to 0.0090 $\mathrm{kg\,m^{-2}\,s^{-1}}$ in eight calls for a moist tropical column), stable columns returned the identity matrix, and all rows summed to one.

2.2 Data

ERA5 (Hersbach et al., 2020): temperature and specific humidity on 32 pressure levels (1000–10 hPa), January, April, July and October, four days per month, 00 and 12 UTC, on a 1° grid; 27 552 columns per year. Tropical Indian Ocean (10°S–10°N, 60–100°E): training 2019–2021, validation 2022, test 2023. Tropical Atlantic (10°S–10°N, 40°W–0°), 2023: held-out region. Versions v4 and v5 assumed a surface pressure of 1013 hPa; v6 uses ERA5 surface pressure (single levels, same dates and grid) and excludes columns whose surface pressure is at or below 1000 hPa, where the lowest level lies below the ground (1.9% of Indian Ocean columns, 10% of Atlantic columns, mostly land). Earlier versions (v1–v3) used synthetic soundings from four climate regimes.

Labels. Because of the scheme's memory, each column gets a random number (0–8) of spin-up calls, which gives a realistic spread of the previous cloud-base mass flux m_b^{old}; the next call provides the labels M and m_b^{new}.

Inputs (69 features): T and q on 32 levels, surface pressure, CAPE, CIN and lifting condensation level pressure from a parcel ascent, and m_b^{old}.

2.3 Models

1. **Gate** (MLP, 2×128): probability that convection is active; a column counts as convective above a threshold of 0.5 (v4, v5) or 0.3 (v6, Section 5).

2. **Cloud top** (MLP, 2×128, 32 classes): highest level touched by the convection. At prediction time the *cautious* cloud top is used: the smallest level k with $P(\mathrm{top} \le k) \ge 0.95$.

3. **Redistribution network** (MLP, 3×256, input: features plus the one-hot cloud top k; output: 32×32 scores z_{ij}). Entries above the cloud top are masked and a row-wise softmax gives

$$\hat{M}_{ij} = \frac{\exp z_{ij}}{\sum_{l \in A_i} \exp z_{il}} \quad (j \in A_i), \qquad A_i = \begin{cases} \{0, \ldots, k\} & i \le k \\ \{i\} & i > k \end{cases} \tag{2}$$

so every row is non-negative and sums to one (mass is conserved exactly) and no air goes above the cloud top.

4. **Cloud-base mass flux** (MLP, 2×64, softplus output): m_b^{new}.

Loss. The redistribution network minimises a row-weighted KL divergence,

$$\mathcal{L}_{\text{KL}} = \sum_i w_i \sum_j M_{ij} \ln \frac{M_{ij}}{\hat{M}_{ij}}, \qquad w_i = 0.8 \, \frac{\Delta p_i (1 - M_{ii})}{\sum_l \Delta p_l (1 - M_{ll})} + 0.2 \, \frac{\Delta p_i}{\sum_l \Delta p_l}, \tag{3}$$

which weights rows by the air FLEXPART actually moves out of them. During training the network receives the true cloud top plus a random 0–3 levels of slack. After 60 epochs, a 10-epoch fine-tuning stage (learning rate 10^{-4}) adds the relative error of the upward and downward mass fluxes F_k across every level interface k,

$$\mathcal{L} = \mathcal{L}_{\text{KL}} + 0.1 \left(\frac{\overline{(\hat{F}^{\uparrow} - F^{\uparrow})^2}}{\overline{(F^{\uparrow})^2}} + \frac{\overline{(\hat{F}^{\downarrow} - F^{\downarrow})^2}}{\overline{(F^{\downarrow})^2}} \right), \qquad F_k^{\uparrow} = \sum_{i \le k} \sum_{j > k} \Delta p_i M_{ij}. \tag{4}$$

2.4 Validation

On the test set: gate F1; cloud-top accuracy; mean absolute matrix error; the worst-level relative error of the mean upward mass-flux profile; the per-column relative error of the off-diagonal part of the matrix (the air that actually moves), $\sum_{i \neq j} |\hat{M}_{ij} - M_{ij}| / \sum_{i \neq j} M_{ij}$, overall and for the 10% of columns with the strongest convection; a particle test (1-D Wasserstein distance between final-pressure distributions of mass-weighted particles); R^2 of m_b^{new}; and speed per column on a CPU.

3 Development history

Table 2 summarises the versions. Two lessons shaped the final design. First, averaged metrics hide structural errors: v2's matrix error was tiny (5.7×10^{-5}) because most entries are trivially near 0 or 1, yet single columns were visibly blurred (Fig. 1a): when unsure whether the cloud stops at one level or another, the network spread air over both. The cloud-top model and masking (v3) removed this hedging and also removed the boundary-layer flux bias. Second, a mass-flux loss switched on from the start collapsed training: the untrained network's fluxes were thousands of times too large, and the quickest way to reduce a *relative* flux error was to move no air at all (Fig. 1b). Used only as a fine-tuning stage (v5), the same term removed the 15–20% flux overestimate seen on ERA5 in v4. v6 added real surface pressure and a lower gate threshold (Section 5).

4 Results (v5)

4.1 Unseen year: tropical Indian Ocean, 2023

On 27 552 test columns (25 359 convective), the gate reaches F1 = 0.989 (24 858 hits, 53 false alarms, 501 misses). The most likely cloud top is exact in 82.7% of convective columns and within one level in 93.7%; the cautious cloud top lies at or above the true one in 99.0% of columns, at a cost of only 0.78 extra levels on average. The mean upward mass-flux profile matches FLEXPART from

Version	Data	Gate F1	Top exact	Flux err.	Off-diag. err.	$R^2(m_b)$	Speed-up
v2	synthetic	0.956	–	10.9%	–	0.964	2.6×
v3, flux loss 0.1	synthetic	0.961	0.845	865%	122%	0.964	1.5×
v3, no flux loss	synthetic	0.961	0.845	4.7%	57%	0.964	1.6×
v4	ERA5	0.989	0.791	28.2%	34%	0.990	1.3×
v5	ERA5	0.989	0.827	10.9%	34%	0.990	1.4×
v6	**ERA5**	**0.994**	**0.817**	**4.7%**	**29%**	**0.991**	**2.4**×*

Table 2: Development history. Flux error: worst-level relative error of the mean upward mass flux. Off-diagonal error: median per-column relative error of the moved air. Synthetic and ERA5 results are not directly comparable; the v6 test set excludes below-ground columns. *FLEXPART's own timing varied between Colab runs (54 vs 103 μs per column), so speed-ups are not comparable between versions. v1 used a simplified stand-in scheme and is omitted.

the surface to the stratosphere (Fig. 2, top), with a worst-level error of 10.9%. The median per-column error of the moved air is 34% (33% for the strongest 10%), the median particle Wasserstein distance is 0.014 hPa, $R^2(m_b^{\text{new}}) = 0.990$, and every row sums to one.

The strongest test column illustrates the cautious cloud top (Fig. 2, bottom): the most likely cloud top was wrong (level 5 instead of 13), which in v4 had wiped out the column, but the cautious top was exactly right, and the emulator reproduces FLEXPART's structure: the surface updraft spread over levels 2–13 and the shallow mixing just above the diagonal.

Real ERA5 convection over the Indian Ocean differs from the synthetic soundings: the upward flux is largest at the surface and decreases steadily with height, and much of the moved air mixes only one to three levels upward.

4.2 Unseen region: tropical Atlantic, 2023

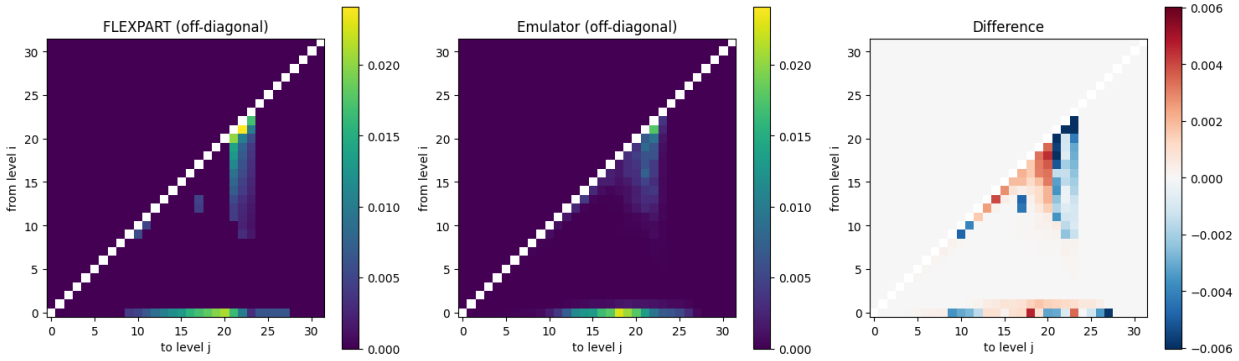
Table 3 compares the two test sets. Every error stays within 1.13 times the Indian Ocean value, meeting the target of 1.5; the mass-flux profile is matched even better (Fig. 3). The off-diagonal error histogram shows a spike at exactly 1.0 (about 800 columns), where the emulator moved no air while FLEXPART did: these are columns where the gate missed weak convection, which also appear as zeros in the m_b^{new} scatter.

Check	Indian Ocean 2023	Atlantic 2023 (unseen)	Ratio
Gate F1	0.989	0.976	–
Cautious top $\geq$ true top	99.0%	98.5%	–
Mass flux, worst level	10.9%	7.9%	0.73
Off-diagonal error, median	33.9%	37.9%	1.12
Off-diagonal error, strongest 10%	33.2%	37.5%	1.13
Particle test, median	0.014 hPa	0.013 hPa	0.93
$R^2(m_b^{\text{new}})$	0.990	0.965	–

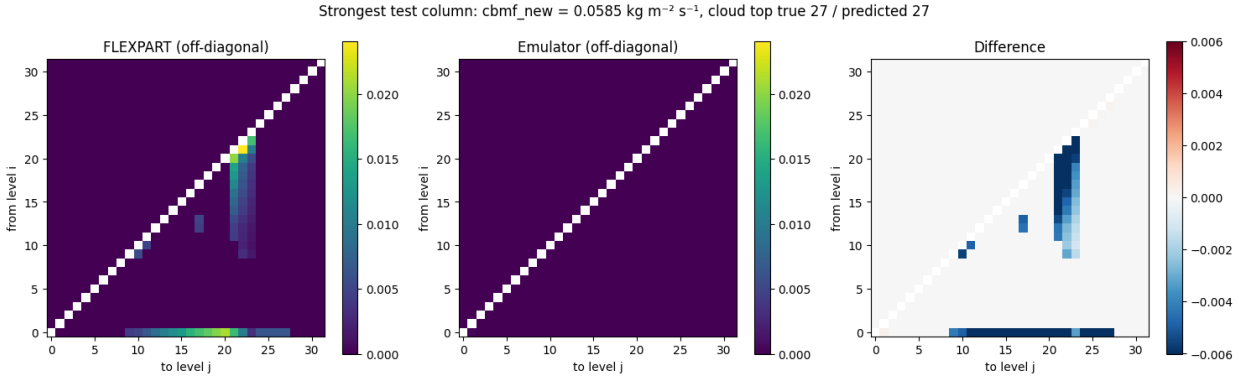
Table 3: Generalisation to a region never seen in training. Convective share: 92% (Indian Ocean test) and 86% (Atlantic).

4.3 Speed

FLEXPART's scheme takes about 50–90 μs per column in these tests (including Python call overhead, and up to about 100 μs on a busier machine); the four-model emulator takes about 38–42 μs per column in batches on a CPU, a speed-up of only 1.3–2.4. Called one column at a time inside FLEXPART, the advantage would shrink further.



(a) v2: the emulator blurs the sharp structure of FLEXPART's matrix.



(b) v3 with a flux loss from the start: the emulator collapses to "no air moves".

Figure 1: Off-diagonal part of the redistribution matrix for the strongest-convection test column (FLEXPART, emulator, difference; synthetic data).

5 Surface pressure, gate threshold and the generalisation gap (v6)

Gate threshold. Re-evaluating the saved v5 model with gate thresholds from 0.5 to 0.2 (no retraining) showed that 91–98% of the missed convective columns are weak convection (the lowest quarter of m_b^{new}). A threshold of 0.2 recovers about 40% of the misses at roughly one extra false alarm each, while the transport checks change by less than 0.3 percentage points. v6 therefore uses 0.3.

Surface pressure. v6 uses ERA5 surface pressure as an input feature (it was constant before) and as the bottom half-level pressure passed to `CONVECT`, as FLEXPART's `calcmatrix` does, and it excludes below-ground columns. Table 4 compares v5 and v6. On the Indian Ocean test year, v6 halves the worst-level mass-flux error (10.9% to 4.7%) and lowers the per-column error (34% to 29%); the gate misses drop from 501 to 154. In the Atlantic, the per-column structure, particle test and m_b^{new} all improve, but the worst-level mass-flux error rises from 7.9% to 11.2%. Part of the improvement may come from excluding land columns, whose lowest levels were extrapolated; this run cannot separate the two effects.

Diagnosing the Atlantic gap. Three findings explain the rise (Table 5). First, the Atlantic's worst level (137.5 hPa) carries only 1% of the peak flux; where the flux is at least 10% of its peak, the worst error is 9.5% (Indian Ocean: 4.6%). Even so, there is a systematic bias: between 860 and 375 hPa the emulator lifts 6–9% too much air in the Atlantic, against 3–4.6% in the Indian Ocean. Second, the Atlantic's surface pressure lies outside the training range for 56% of its columns (median 1013.8 against 1009.4 hPa), and those columns have a much larger error (28.6% against 9.8%). Third, clipping the surface-pressure *input* to the training range barely changes the error (12.3% to 11.9%), so the input itself is not the cause. Higher surface pressure marks a different regime: Atlantic convection is shallower (its upward flux falls to 0.16 hPa at 837.5 hPa, against 0.27 hPa in the Indian Ocean), which the Indian Ocean training data rarely contains, so the

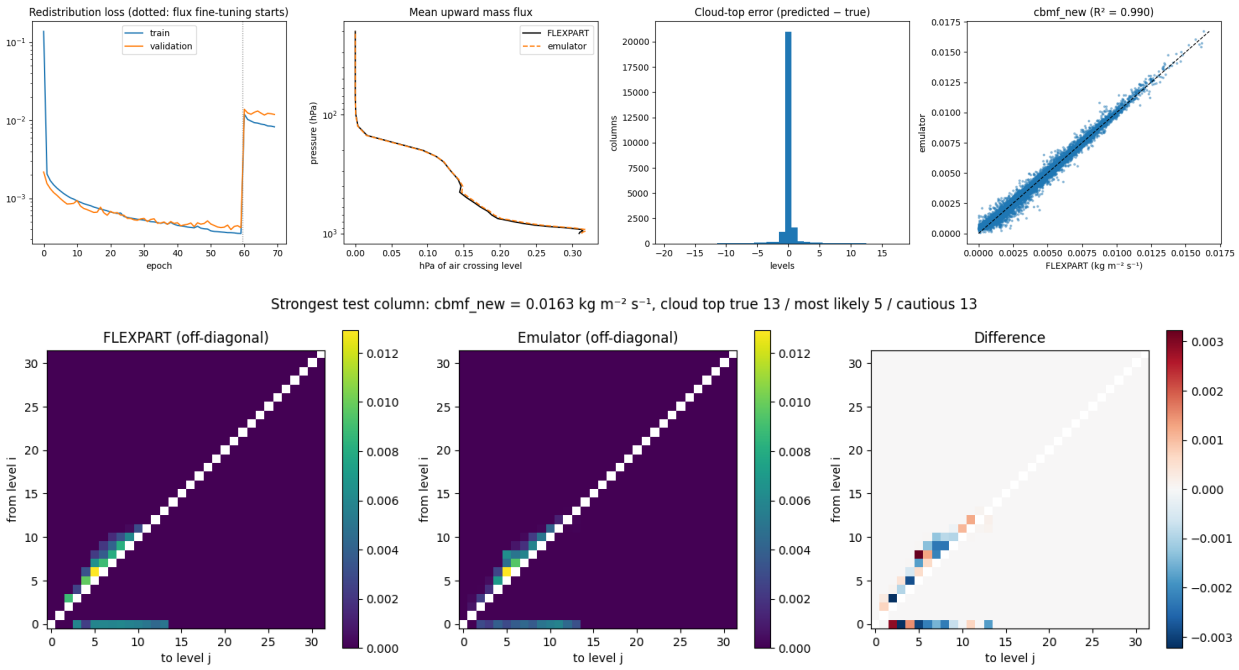


Figure 2: v5 on the Indian Ocean test year 2023. Top: training loss (dotted line: start of flux fine-tuning), mean upward mass flux, cloud-top error and m_b^{new}. Bottom: off-diagonal matrix of the strongest test column.

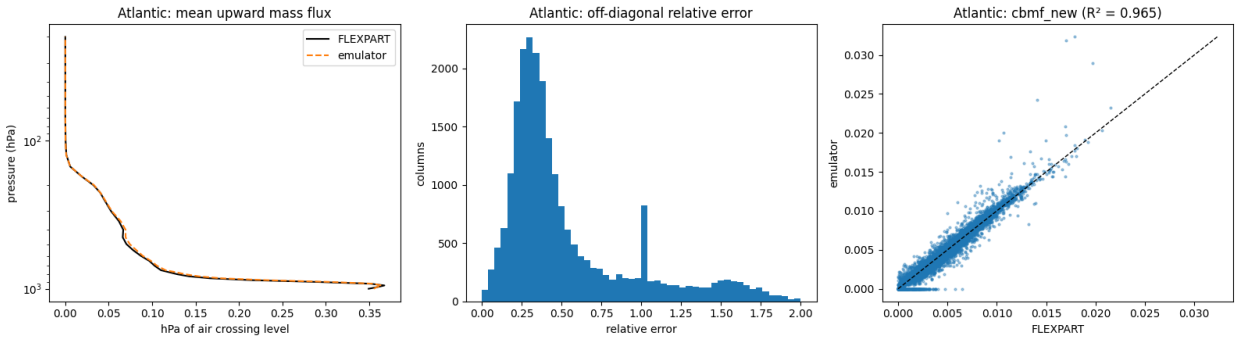


Figure 3: v5 on the tropical Atlantic 2023 (held-out region): mean upward mass flux, per-column off-diagonal error and m_b^{new}.

emulator treats it as slightly deeper than it is. The remedy is more varied training data rather than a different model.

6 Discussion

What worked. Extracting the real scheme made true labels cheap (about 0.2 ms per call). A masked, row-wise softmax conserves mass exactly by construction. The cloud-top model, used cautiously, removed the blurring and made cloud-top mistakes cheap. Weighting rows by moved air and a late flux fine-tuning stage brought the mass-flux profile into agreement. Real surface pressure (v6) halved the remaining Indian Ocean flux error.

What did not. A flux loss from the start collapsed training. Speed was the weakest argument: the Emanuel scheme is cheaper than expected. Training on one region left a regime gap: the mass-flux profile transfers less well to a region with shallower convection.

Limitations. Offline only; below-ground (mostly land) columns excluded; ERA5 pressure levels instead of FLEXPART's model levels; a 1° grid, one training region and one held-out region; and FLEXPART's `nconvtop` lies one level above the highest level with mass flux, which may place

Check	v5 Indian	v6 Indian	v5 Atlantic	v6 Atlantic
Gate F1	0.989	0.994	0.976	0.981
Missed / false alarms	501 / 53	154 / 148	643 / 503	350 / 499
Mass flux, worst significant level	10.9%	4.7%	7.9%	11.2%
Off-diagonal error, median	33.9%	29.3%	37.9%	32.8%
Off-diagonal error, strongest 10%	33.2%	29.3%	37.5%	30.3%
Particle test, median	0.014 hPa	0.013 hPa	0.013 hPa	0.011 hPa
$R^2(m_b^{\text{new}})$	0.990	0.991	0.965	0.978

Table 4: v5 against v6 on the two test sets (2023). The v6 test sets exclude below-ground columns.

	Indian Ocean 2023	Atlantic 2023
Surface pressure, 5 / 50 / 95% (hPa)	1006.5 / 1010.6 / 1013.4	1010.6 / 1013.8 / 1017.1
Outside training range (1004.0–1013.5 hPa)	5.0%	56.2%
Worst-level flux error (level; share of peak)	4.6% (837.5; 79%)	12.3% (137.5; 1%)
Worst error where flux $\geq 10\%$ of peak	4.6%	9.5%
Error inside / outside training range	–	9.8% / 28.6%
Error, surface-pressure input clipped	–	11.9% (unclipped 12.3%)

Table 5: Diagnostic of the v6 mass-flux error. Training surface pressure (2019–2021): 5 / 50 / 95% = 1006.2 / 1009.4 / 1012.1 hPa. Labels were regenerated with a new random spin-up, hence 12.3% instead of 11.2%.

cloud tops slightly high.

Next steps. (1) Train on several regions with different convection regimes (e.g. adding the Atlantic and testing on the western Pacific) and give weak convection more weight in the gate; (2) retrain on FLEXPART's terrain-following model levels, so land columns can be kept; (3) Phase 2: call the TorchScript model from FLEXPART's Fortran code via FTorch and run long simulations; (4) Phase 3: train turbulence or scavenging modules on LES or observations, where an ML module could be more accurate than the original empirical fits.

Code and data

Notebooks, scripts, results and figures: this repository. FLEXPART is downloaded from its official repository at run time; ERA5 data from the Copernicus Climate Data Store.

References

Bakels, L., et al. (2024). Version 11 of the Lagrangian particle dispersion model FLEXPART. *Geosci. Model Dev.*, 17.

Emanuel, K. A. and Živković-Rothman, M. (1999). Development and evaluation of a convection scheme for use in climate models. *J. Atmos. Sci.*, 56, 1766–1782.

Fillola, E., et al. (2023). A machine learning emulator for Lagrangian particle dispersion model footprints: a case study using NAME. *Geosci. Model Dev.*, 16, 1997.

Forster, C., Stohl, A. and Seibert, P. (2007). Parameterization of convective transport in a Lagrangian particle dispersion model and its evaluation. *J. Appl. Meteor. Climatol.*, 46, 403–422.

He, T.-L., et al. (2025). FootNet v1.0: development of a machine learning emulator of atmospheric transport. *Geosci. Model Dev.*, 18, 1661.

Hersbach, H., et al. (2020). The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.*, 146, 1999–2049.

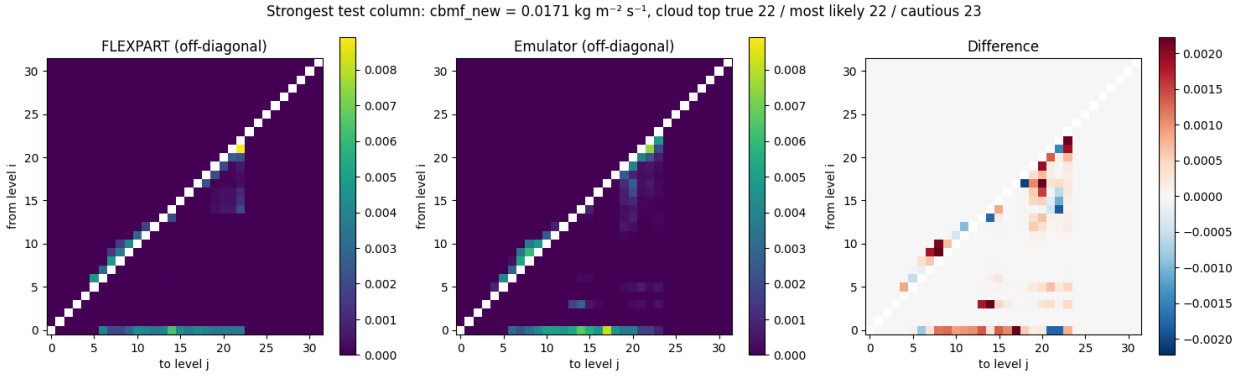


Figure 4: v6, strongest test column of the Indian Ocean 2023 (a deep column; cloud top true 22, most likely 22, cautious 23). The emulator reproduces the surface updraft to levels 6–22 and the mixing above the diagonal; surface air detrains slightly too low.

Pisso, I., et al. (2019). The Lagrangian particle dispersion model FLEXPART version 10.4. *Geosci. Model Dev.*, 12, 4955–4997.

Rasp, S., Pritchard, M. S. and Gentine, P. (2018). Deep learning to represent subgrid processes in climate models. *PNAS*, 115, 9684–9689.

Stohl, A., et al. (2005). Technical note: The Lagrangian particle dispersion model FLEXPART version 6.2. *Atmos. Chem. Phys.*, 5, 2461–2474.